



BANNARI AMMAN INSTITUTE OF TECHNOLOGY

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Lecture Notes

22GE004 – BASICS OF ELETRONICS ENGINEERING

UNIT 4 - OVERVIEW OF LOGIC GATES

Introduction

The digital world relies on a fundamental concept: binary data. Information is processed and stored using just two values, typically represented as 0 (LOW voltage) and 1 (HIGH voltage). Logic gates are the building blocks of digital circuits. These electronic circuits perform basic logical operations on binary inputs and produce a single binary output. They are like tiny decision-makers, following specific rules based on the type of gate and the input they receive.

Understanding Logic Gates

- **Function:** Logic gates perform basic logical operations like AND, OR, and NOT on binary inputs.
- **Inputs and Outputs:** Each gate has one or more input lines (usually 1, 2, or 3) and a single output line.
- **Operation:** The output of a logic gate depends on the combination of its inputs and the specific logical function it performs.
- **Representation:** Logic gates are represented by symbols that depict their function.

Basic Types of Logic Gates

1. NOT Gate (Inverter):

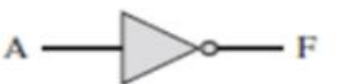
- Performs logical negation (inversion).
- Output is the opposite of the single input.
- Symbol: A circle with a small line entering it.

2. AND Gate:

- Performs logical conjunction (AND).
- Output is 1 only if both inputs are 1.
- Symbol: Two lines entering a dot.

3. OR Gate:

- Performs logical disjunction (OR).
- Output is 1 if at least one input is 1.
- Symbol: Two lines entering a vertical bar.

Name	Graphical Symbol	Algebraic Function	Truth Table															
AND		$F = A \bullet B$ or $F = AB$	<table><tr><th>A</th><th>B</th><th>F</th></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td></tr></table>	A	B	F	0	0	0	0	1	0	1	0	0	1	1	1
A	B	F																
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1	0	0																
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OR		$F = A + B$	<table><tr><th>A</th><th>B</th><th>F</th></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td></tr></table>	A	B	F	0	0	0	0	1	1	1	0	1	1	1	1
A	B	F																
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NOT		$F = \overline{A}$ or $F = A'$	<table><tr><th>A</th><th>F</th></tr><tr><td>0</td><td>1</td></tr><tr><td>1</td><td>0</td></tr></table>	A	F	0	1	1	0									
A	F																	
0	1																	
1	0																	
NAND		$F = \overline{AB}$	<table><tr><th>A</th><th>B</th><th>F</th></tr><tr><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td></tr></table>	A	B	F	0	0	1	0	1	1	1	0	1	1	1	0
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NOR		$F = \overline{A + B}$	<table><tr><th>A</th><th>B</th><th>F</th></tr><tr><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td></tr></table>	A	B	F	0	0	1	0	1	0	1	0	0	1	1	0
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XOR		$F = A \oplus B$	<table><tr><th>A</th><th>B</th><th>F</th></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td></tr></table>	A	B	F	0	0	0	0	1	1	1	0	1	1	1	0
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Additional Gates (Derived from Basics)

- **NAND Gate (NOT-AND):** Acts like an AND gate with an inverted output (output is 1 only if both inputs are 0).
- **NOR Gate (NOT-OR):** Acts like an OR gate with an inverted output (output is 1 only if both inputs are 0).

Summary

Logic gates are the essential building blocks of digital circuits. By understanding their basic functions (AND, OR, NOT) and how they operate on binary inputs, you can grasp the fundamental operations within digital systems like computers and communication devices. These basic gates can be combined to create more complex logic circuits that perform various tasks.